Abdul Wahab

Applied AI Solutions Engineer | Production RAG, LLM Agents & Computer Vision

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LinkedIn | GitHub

Summary

Applied AI Engineer specializing in building production-ready retrieval-augmented generation (RAG) systems, LLM agents, document-intelligence workflows, and highly accurate computer vision pipelines. Experienced in hybrid retrieval, multimodal document processing, API development, and deployment of scalable, business-facing AI architectures that reduce manual overhead, optimize system latency, and deliver measurable ROI.

Experience

Applied AI Engineer *Mar 2026 – Present*

Nexonic (Evyol Group) *Multan, Pakistan*

- Engineered a hybrid-search RAG sales assistant using vector retrieval, BM25, and re-ranking; reduced irrelevant retrievals by approximately **35%** and decreased average response time from **6 to 3 seconds**.
- Developed a supplier-negotiation agent utilizing LangChain and Ollama that automated four pricing and procurement workflows, saving an estimated **20–30 minutes per negotiation**.
- Fine-tuned YOLO models on custom datasets to build a robust Number Plate Recognition (NPR) system deployed across factory and residential sites, achieving **80% mAP@50** for automated access control.
- Designed a multi-stage fire-detection pipeline combining object tracking, temporal evidence, and vision-language-model verification; reduced false positives by **60%** for industrial clients.
- Developed an AI plant disease detection and prevention system for wheat, cotton, and rice crops to support Evyol Group's agricultural division with diagnostic and recommendation pipelines.

Research & Evaluation Engineer - Agricultural AI *Mar 2025 – Sep 2025*

MNS University of Agriculture *Multan, Pakistan*

- Prepared agricultural image data through annotation, preprocessing, and augmentation for large-scale machine-learning experiments.
- Trained and evaluated segmentation and classification models (**70% IoU** on U-Net; **95% accuracy** on ResNet50), gaining experience in dataset quality, reproducible evaluation, and AI integrations.

Selected Projects

Tawreeq – AI Procurement Intelligence Platform GitHub

- Built a full-stack AI procurement platform that ingests supplier quotations and procurement documents, performs OCR-based extraction, and supports bilingual Arabic/English comparison.
- Engineered an end-to-end procurement-intelligence workflow spanning document ingestion, structured extraction, quotation comparison, value analysis, and Excel export for GCC procurement teams.
- **Tech:** Python, FastAPI, React, Supabase/PostgreSQL, Mistral OCR, Docker, AWS ECS.

Neuro-Critics AI – Multimodal Research Critique Agent GitHub

- Developed an AI assistant for critical analysis of neuroscience research, combining a curated knowledge base with paper-specific multimodal reasoning.
- Built a knowledge base of more than 50 neuroscience papers with dynamic ingestion, hybrid search, re-ranking, multimodal LLM analysis, and OCR for figures and diagrams.
- Designed critique workflows to identify methodological weaknesses, statistical limitations, and research gaps while interpreting complex visual content.
- **Tech:** Python, LangChain, FAISS, Streamlit, Hugging Face Embeddings, Gemini API.

Zarai Radar – AI Crop Risk Intelligence Platform GitHub

- Built a crop-risk intelligence platform that delivers location- and growth-stage-aware climate, disease, and pest-risk alerts for major crops.
- Combined rule-based agronomic models, disease detection, and RAG over vectorized agricultural knowledge to produce explainable, actionable crop advisories.
- **Tech:** Python, FastAPI, React, Supabase/pgvector, Hugging Face Embeddings, Weather APIs.

Research & Publications

Cross-Family Benchmark of Plant Disease Detection Models *Manuscript PDF*

Manuscript submitted to the International Journal of Machine Learning and Cybernetics

- Designed and executed a benchmark comparing YOLO26-L, RT-DETR-L, and RF-DETR-L across public rice-leaf and cotton-disease datasets.
- Ran six detector–dataset configurations and 18 training runs across three random seeds under harmonized evaluation settings.
- Evaluated mAP, precision, recall, F1, inference speed, and run-to-run variability to assess accuracy, robustness, and deployment efficiency.
- Found that RF-DETR-L achieved the strongest mAP/F1 results, while YOLO26-L delivered the highest throughput, revealing an accuracy–efficiency trade-off.
- Applied statistical comparison and cross-dataset analysis to examine changes in detector rankings across datasets.

Technical Skills

AI Systems & Generative AI: LLM Applications, Retrieval-Augmented Generation (RAG), AI Agents, Tool Calling, Workflow Orchestration, LangChain, LangGraph, LLM Evaluation & Tracking, Langfuse.
Retrieval & Data Systems: Hybrid Search, BM25, Dense Retrieval, Cross-Encoder Reranking, FAISS, Qdrant, pgvector, PostgreSQL, MySQL, Supabase, Neo4j.
Backend & Application Engineering: Python, FastAPI, REST APIs, SQL, Redis, Asynchronous Processing.
Machine Learning & Computer Vision: PyTorch, TensorFlow/Keras, OpenCV, YOLO, ResNet, U-Net, Object Detection, Model Benchmarking and Evaluation, Detection Transformers.
Infrastructure & Developer Tools: Docker, Git, GitHub, AWS, CI/CD.

Education & Leadership

Bachelor of Science in Computer Science *Sep 2023 – Sep 2027*

MNS-University of Agriculture | CGPA: 3.76/4.00 *Multan, Pakistan*

- **Section Leader, Stanford Code in Place (2026):** Globally selected to mentor 20+ international students.
- **1st Place**, Speed Programming Competition, BZU Multan (2025) | **2nd Place**, ACM-W Air University (2025).